PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: StockSage

Version: 1.0

1. Project Overview

- **Problem Statement:** In Malaysia and the rest of the world, many individuals, businesses, and communities face challenges in making timely and informed decisions due to fragmented data, limited analytical capabilities, and reliance on manual or intuition-based processes. Participants are required to build an AI-powered decision intelligence system, where Z.AI's GLM (General Language Model) plays a central role in analysing information, generating insights, and supporting decision-making. The solution must demonstrate how Z.AI's GLM enables the following points.

- **Target Domain:** AI for Economic Empowerment & Decision Intelligence

- **Proposed Solution Summary:** StockSage is an AI-powered retail advisor designed for SME grocery stores in Malaysia. Instead of only tracking stock, StockSage analyses expiry dates, stock levels, sales trends, and external factors such as weather or holidays to recommend profit-driven actions such as promotions, bundle deals, or discount timing. The system transforms ordinary inventory data into smart business decisions that reduce wastage and increase revenue.

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2. Background & Business Objective

- **Background of the Problem:** Many small retail stores lose money every month due to expired products, poor inventory turnover, and reactive management. Existing systems notify owners when stock is low or expired, but they do not recommend what action should be taken.

  For example, if 20 cartons of milk are expiring in 2 days, most systems only show an alert. Store owners still need to think manually about pricing strategy, promotions, and timing.

- **Importance of Solving This Issue:** Reducing product wastage directly improves profit margins for SMEs. Smart recommendations also help store owners make faster and better decisions without needing business analytics expertise.

- **Strategic Fit / Impact:** StockSage aligns strongly with the hackathon theme by using AI as a core decision-making engine. Z.AI GLM transforms structured and unstructured data into understandable business actions, helping SMEs become more competitive and sustainable.

3. Product Purpose

3.1. Main Goal of the System: To become an AI business advisor for SME retailers that helps reduce waste, optimize promotions, and maximize profits through intelligent recommendations.

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3.2. Intended Users (Target Audience):

- SME Business Owners
- Mini Market Operators
- Convenience Store Managers

4. System Functionalities

4.1. Description: The system operates through three intelligent layers:

Data Layer (Oracle SQL)

Stores product details, stock quantities, expiry dates, and historical sales records.

Decision Layer (Spring Boot + Z.AI GLM)

Fetches business data, sends context to GLM, receives strategic recommendations, and calculates potential outcomes.

User Layer (Frontend Dashboard)

Displays critical stock alerts, AI insights, and suggested actions in a simple dashboard.

4.2. Key Functionalities:

- **Smart Inventory Monitoring:** Displays all products with stock quantity, expiry dates, and risk status labels such as Safe, Warning, and Critical.

- **AI Decision Recommendation Engine:** Uses Z.AI GLM to analyse inventory risk and generate actionable strategies such as promotions, stock clearance campaigns, or urgent prioritisation.

- **Editable Inventory Management:** Allows users to manually update product quantity and expiry dates directly from the dashboard to keep records accurate.

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- **Profit Risk & Savings Analytics:** Calculates current stock value, profit at risk from expiring products, and estimated savings if AI recommendations are executed.

- **One-Click Promotion Execution:** Users can instantly activate AI-generated promotional actions through the "Execute Promotion" button.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI GLM is selected because it can perform reasoning, summarisation, recommendation generation, and explanation. It is ideal for converting raw retail data into actionable business strategies.

4.3.2. **Prompting Strategy:**

We use multi-step prompting:

Step 1: Summarise inventory risk
Step 2: Analyse historical sales trend
Step 3: Generate best promotion strategy
Step 4: Explain reasoning clearly

This method ensures better and more reliable outputs.

4.3.3. **Context & Input Handling**:

The system accepts:

- Structured data (SQL records)
- Semi-structured data (CSV uploads)
- External signals (holiday/weather)

Large datasets are chunked by product category or urgency level before sending to the model.

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4.3.4. **Fallback & Failure Behavior:** If AI output is unclear or irrelevant:

- Retry with simplified prompt
- Use rule-based fallback suggestions
- Show safe default recommendation
- Display "AI confidence low" status to user

5. User Stories & Use Cases

- **User Stories:** As a store owner, I want to know which products are at risk of expiry so I can act early.
  As a manager, I want AI to recommend promotions so I can increase sales quickly.
  As a business owner, I want to see estimated savings so I know whether to follow the suggestion.

- **Use Cases (Main Interactions):**

  Inventory Risk Detection Flow: User opens dashboard, then system retrieves product data from Oracle SQL, Products are classified into Safe / Warning / Critical categories and high-risk products are highlighted.

  AI Recommendation Flow: System detects critical products nearing expiry then sends stock and risk data to Z.AI GLM. AI returns promotional strategy and reasoning then the dashboard displays Intelligent Action Plan.

  Inventory Update Flow: User edits quantity or expiry date and clicks Save Changes then updated data will be stored in database, the dashboard will refresh with latest risk analysis.

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Promotion Execution Flow: User reviews AI suggestion and clicks Execute Promotion, then promotion is triggered for store action / POS campaign simulation

6. Features Included (Scope Definition)

- Real-time inventory dashboard with stock health monitoring
- Product risk categorisation (Safe, Warning, Critical)
- AI-generated promotional and clearance recommendations
- Logic & insight explanation for every AI decision
- Current stock value calculation
- Profit at risk detection
- Estimated savings projection
- Editable stock quantity and expiry records
- One-click promotion execution simulation
- Oracle SQL + Spring Boot + Frontend integration

7. Features Not Included (Scope Control)

- Direct payment gateway integration
- Supplier ordering automation
- Multi-branch synchronization (initial version)
- Real-time IoT shelf sensors

8. Assumptions & Constraints

- **LLM Cost Constraint:** to reduce token usage;
    - Only risky products are sent to AI
    - Frequent recommendations are cached
    - Summarised data sent instead of full database tables

  Estimated cost per session remains low and scalable.
- **Technical Constraints:**
    - Requires internet access for GLM calls

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  - Oracle DB setup required
  - MVP supports limited product categories
- **Performance Constraints:** Response time may slow if too many products are analysed simultaneously.
- **User Input:** Human approval is required before any promotion is launched.

9. Risks & Questions Throughout Development

- **Recommendation Accuracy:** Will suggested promotions always produce positive profit impact?
- **Data Quality:** What happens if expiry dates or stock records are inaccurate?
- **User Trust:** Will store owners trust AI recommendations enough to take action?
- **External Factors:** Unexpected events may reduce effectiveness of suggested campaigns.
- **Scalability:** Can the system support multiple stores in future versions?

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